Petra Zlatarić-Novosel

Machine Learning Engineer | NLP and Document AI

PROFESSIONAL SUMMARY

Machine Learning Engineer focused on NLP and document AI, with experience building and deploying text classification and named entity recognition systems for Croatian documents. Production work includes a document classifier serving approximately 15,000 documents per day and model drift monitoring. Experienced with Python, Azure, FastAPI, Docker, PyTorch, Hugging Face Transformers, and MLflow.

SKILLS

Programming and data: Python, pandas, Oracle SQL, PostgreSQL
Machine learning and NLP: PyTorch, Hugging Face Transformers, spaCy, scikit-learn, fastText, sentence-transformers, FAISS
Production and MLOps: FastAPI, Docker, Azure App Service, Azure ML, Azure Functions, MLflow, Evidently

EXPERIENCE

ML Engineer | Slavonica Document AI | 2023–Present

- Built and deployed a classifier for scanned incoming documents across 23 classes, achieving 94% accuracy and a macro F1 score of 0.91.
- Fine-tuned XLM-R base and served the model as a FastAPI service in Docker on Azure App Service, processing approximately 15,000 documents per day.
- Developed a Croatian named entity recognition model for persons, organisations, OIB numbers, addresses, and legal references in administrative text, fine-tuning BERTic.
- Built an annotated dataset of 4,000 documents with an annotation guideline and two annotators; inter-annotator kappa was 0.87.
- Implemented weekly model drift reports for input text length, vocabulary, and predicted class distribution, with alerts when PSI exceeded 0.2. Detected a silent OCR change that reduced accuracy by 9 points; the issue was fixed within a week.
- Published NER work at a regional workshop in 2024.

Data Scientist | Baranjska kreditna banka | 2021–2023

- Developed models for customer churn prediction and complaint classification.
- Built a customer complaint topic classifier using a TF-IDF and logistic regression baseline followed by fastText, covering 11 topics for weekly reporting to customer service management.

PROJECTS

Retrieval-Augmented Question Answering Demo | Personal project | 2025

- Built a weekend prototype for question answering over Croatian laws using a sentence embedding model and an open-weight LLM through an API. The prototype was not used in production and was checked manually.

EDUCATION

Doctoral studies in progress | University of Drava | Expected 2027
Research topic: Named entity recognition in Croatian legal texts.

CERTIFICATION AND TEACHING

Microsoft Certified: Azure AI Engineer Associate | 2024
Guest lecture, NLP Summer School | 2025 | 90 minutes